

CSA0910 - PROGRAMMING IN JAVA

CLASS ASSIGNMENT – 01

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1. Write Five Advantages of Java

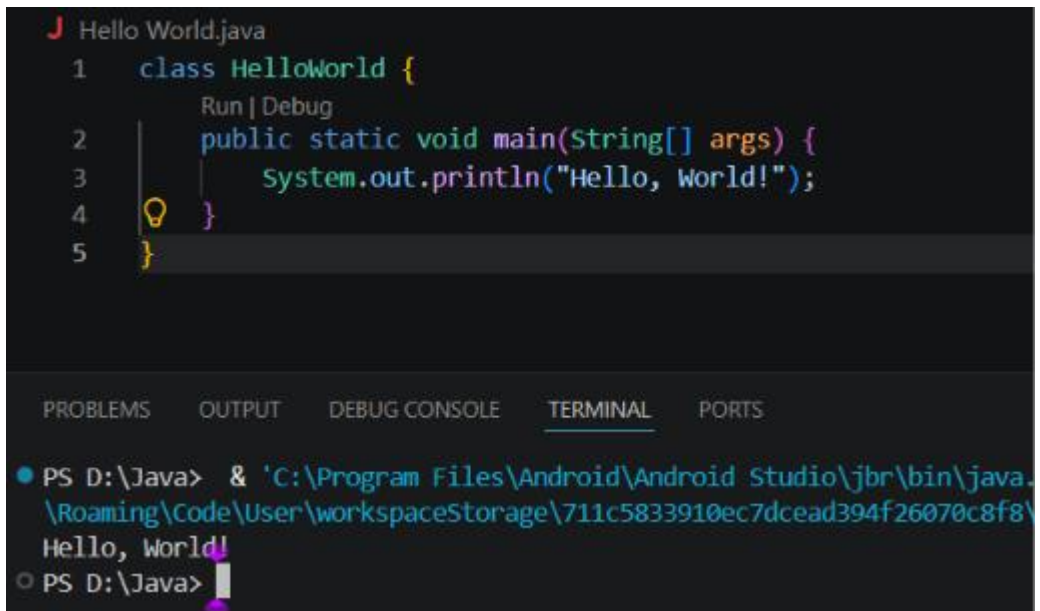
1. Platform Independent – Java programs run on any operating system using the JVM.
2. Object-Oriented – Supports concepts like classes, objects, inheritance, and polymorphism.
3. Secure – Provides built-in security features such as bytecode verification.
4. Robust – Strong memory management and exception handling reduce errors.
5. Multithreaded – Allows multiple tasks to execute simultaneously.

2. Compare Java with C/C++

Java	C/C++
Platform independent	Platform dependent
Uses JVM	Compiles directly to machine code
Object-oriented	C is procedural, C++ is object-oriented
Automatic garbage collection	Manual memory management
More secure	Less secure

3. Hello World Program

```
class HelloWorld {  
    public static void main(String[] args) {  
        System.out.println("Hello World");  
    }  
}
```



The screenshot shows an IDE window titled "Hello World.java" with the following code:

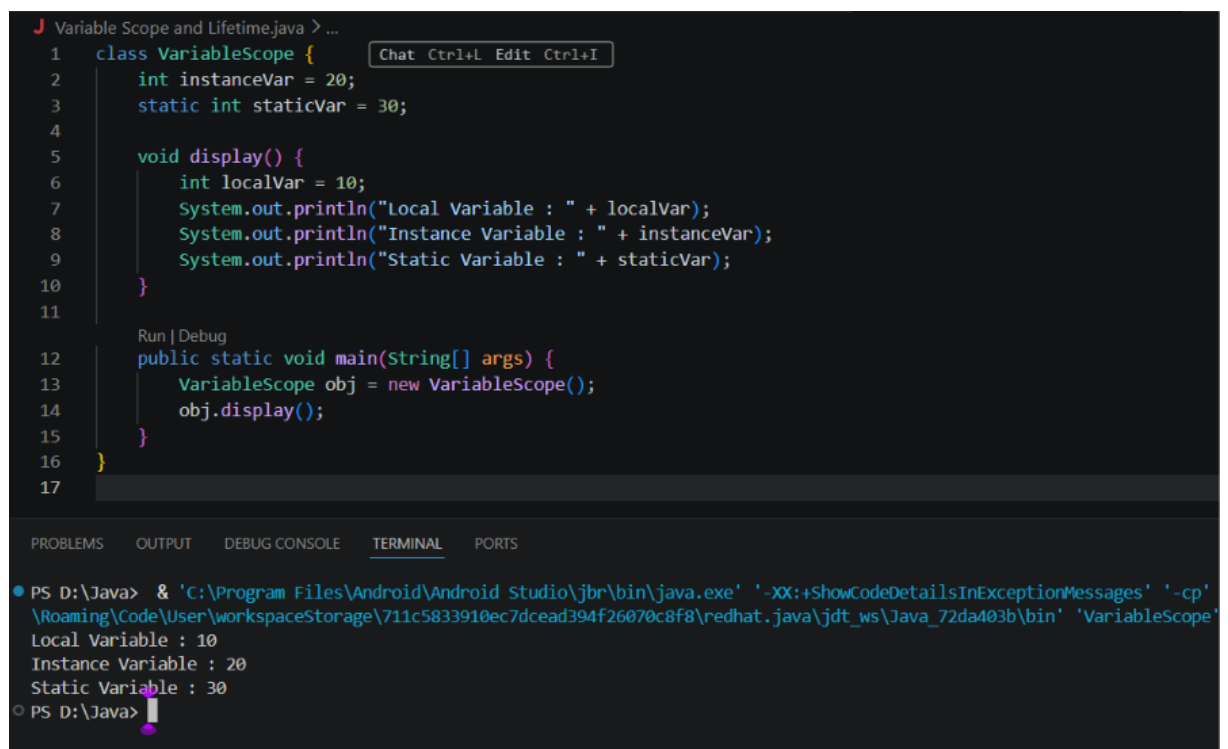
```
1 class HelloWorld {  
2     public static void main(String[] args) {  
3         System.out.println("Hello, World!");  
4     }  
5 }
```

Below the code editor, the "TERMINAL" tab is active, showing the command prompt output:

```
PS D:\Java> & 'C:\Program Files\Android\Android Studio\jbr\bin\java.  
Hello, World!  
PS D:\Java>
```

4. Program Demonstrating Variable Scope and Lifetime

```
class VariableScope {  
    int instanceVar = 20;  
    static int staticVar = 30;  
  
    void display() {  
        int localVar = 10;  
        System.out.println("Local Variable : " + localVar);  
        System.out.println("Instance Variable : " + instanceVar);  
        System.out.println("Static Variable : " + staticVar);  
    }  
  
    public static void main(String[] args) {  
        VariableScope obj = new VariableScope();  
        obj.display();  
    }  
}
```

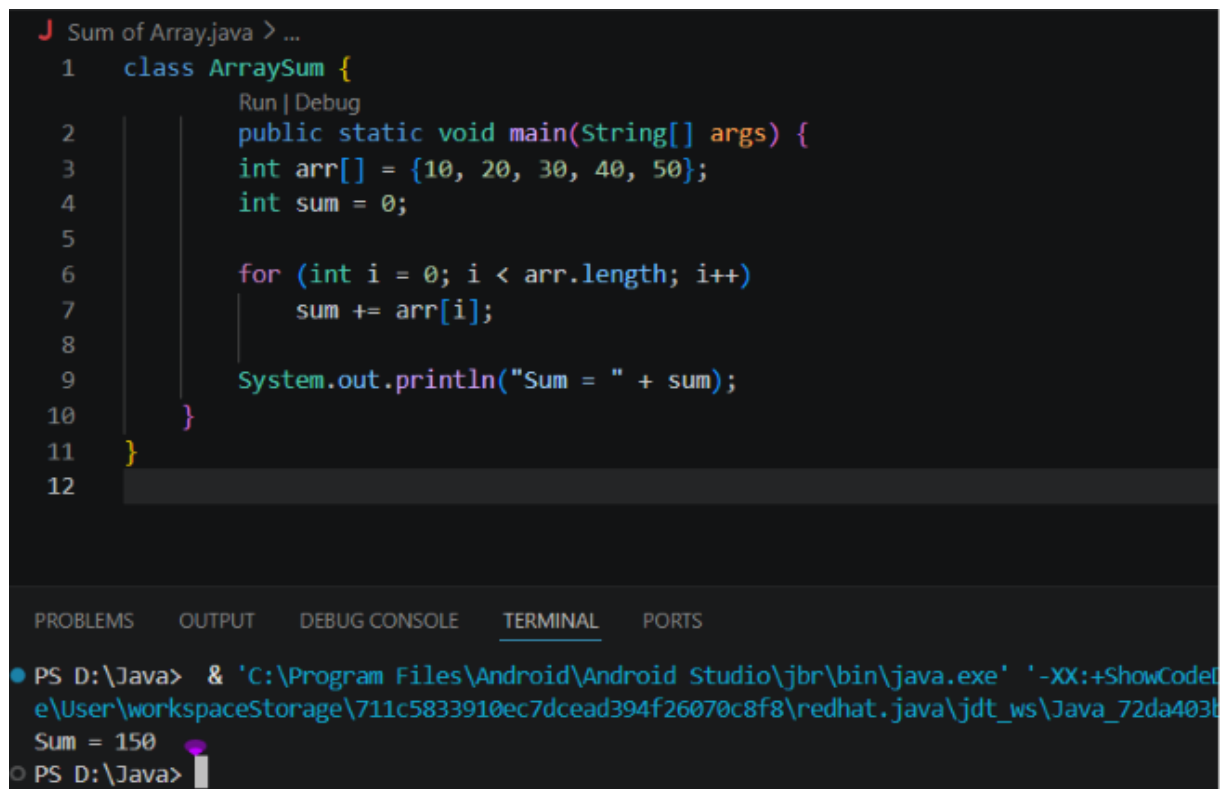


The screenshot shows an IDE window titled "Variable Scope and Lifetime.java > ...". The code editor displays the same Java code as shown in the previous block. Below the code editor, there is a "Run | Debug" button. The "TERMINAL" tab is active, showing the command prompt output of the program execution. The output is as follows:

```
PS D:\Java> & 'C:\Program Files\Android\Android Studio\jbr\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' '  
\\Roaming\Code\User\workspaceStorage\711c5833910ec7dcead394f26070c8f8\redhat.java\jdt_ws\Java_72da403b\bin' 'VariableScope'  
Local Variable : 10  
Instance Variable : 20  
Static Variable : 30  
PS D:\Java>
```

5. Program to Find Sum of Array Elements

```
class ArraySum {  
    public static void main(String[] args) {  
        int arr[] = {10, 20, 30, 40, 50};  
        int sum = 0;  
  
        for (int i = 0; i < arr.length; i++)  
            sum += arr[i];  
  
        System.out.println("Sum = " + sum);  
    }  
}
```



The screenshot displays an IDE window titled "Sum of Array.java > ...". The code editor shows the following Java code:

```
1  class ArraySum {  
    Run | Debug  
2      public static void main(String[] args) {  
3          int arr[] = {10, 20, 30, 40, 50};  
4          int sum = 0;  
5  
6          for (int i = 0; i < arr.length; i++)  
7              sum += arr[i];  
8  
9          System.out.println("Sum = " + sum);  
10     }  
11 }  
12
```

Below the code editor, the "TERMINAL" tab is active, showing the command prompt output:

```
PS D:\Java> & 'C:\Program Files\Android\Android Studio\jbr\bin\java.exe' '-XX:+ShowCodeDetails\User\workspaceStorage\711c5833910ec7dcead394f26070c8f8\redhat.java\jdt_ws\Java_72da403b  
Sum = 150  
PS D:\Java>
```

6. Program to Find Average of Array Elements

```
class ArrayAverage {  
    public static void main(String[] args) {  
        int arr[] = {10, 20, 30, 40, 50};  
        int sum = 0;  
  
        for (int i = 0; i < arr.length; i++)  
            sum += arr[i];  
  
        double avg = (double) sum / arr.length;  
  
        System.out.println("Average = " + avg);  
    }  
}
```



The screenshot shows an IDE with a Java file named 'Average of Array.java'. The code is as follows:

```
1 class ArrayAverage {  
2     public static void main(String[] args) {  
3         int arr[] = {10, 20, 30, 40, 50};  
4         int sum = 0;  
5  
6         for (int i = 0; i < arr.length; i++)  
7             sum += arr[i];  
8  
9         double avg = (double) sum / arr.length;  
10  
11        System.out.println("Average = " + avg);  
12    }  
13 }
```

The IDE interface includes a 'Run | Debug' button and a 'Terminal' tab. The terminal output shows the command to run the program and the resulting average value:

```
PS D:\Java> & 'C:\Program Files\Android\Android Studio\jbr\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Sae\User\workspaceStorage\711c5833910ec7dcead394f26070c8f8\redhat.java\jdt_ws\Java_72da403b\bin' 'ArrayAverage'  
Average = 30.0  
PS D:\Java>
```

7. Program to Reverse an Array

```
class ArrayReverse {  
    public static void main(String[] args) {  
        int arr[] = {10, 20, 30, 40, 50};  
  
        System.out.println("Reversed Array:");  
  
        for (int i = arr.length - 1; i >= 0; i--)  
            System.out.print(arr[i] + " ");  
    }  
}
```



The screenshot shows an IDE with a Java file named 'Reverse an Array.java'. The code is as follows:

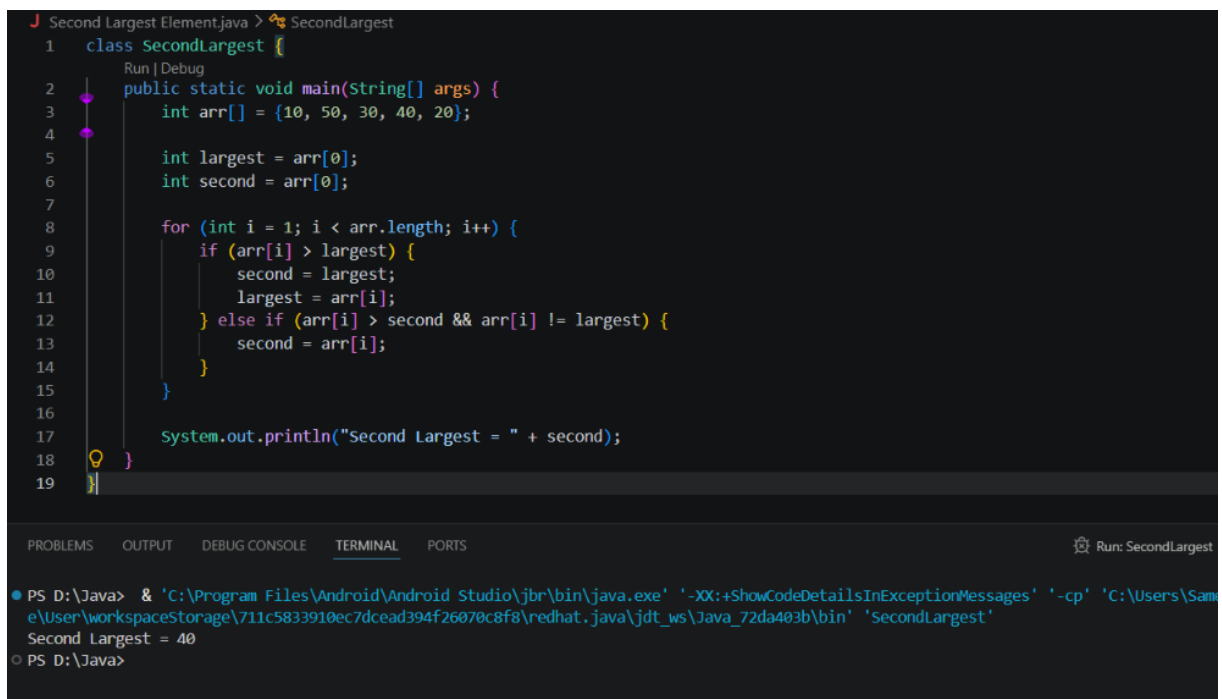
```
1 class ArrayReverse {  
2     public static void main(String[] args) {  
3         int arr[] = {10, 20, 30, 40, 50};  
4  
5         System.out.println("Reversed Array:");  
6  
7         for (int i = arr.length - 1; i >= 0; i--)  
8             System.out.print(arr[i] + " ");  
9     }  
10 }
```

The terminal output shows the execution of the program:

```
PS D:\Java> & 'C:\Program Files\Android\Android Studio\jbr\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Sameer G\workspaceStorage\711c5833910ec7dcead394f26070c8f8\redhat.java\jdt_ws\Java_72da403b\bin' 'ArrayReverse'  
Reversed Array:  
50 40 30 20 10  
PS D:\Java>
```

8. Program to Find the Second Largest Element

```
class SecondLargest {  
    public static void main(String[] args) {  
        int arr[] = {10, 50, 30, 40, 20};  
  
        int largest = arr[0];  
        int second = arr[0];  
  
        for (int i = 1; i < arr.length; i++) {  
            if (arr[i] > largest) {  
                second = largest;  
                largest = arr[i];  
            } else if (arr[i] > second && arr[i] != largest) {  
                second = arr[i];  
            }  
        }  
  
        System.out.println("Second Largest = " + second);  
    }  
}
```



The screenshot shows an IDE with a Java file named 'SecondLargest.java'. The code is as follows:

```
1 class SecondLargest {  
2     public static void main(String[] args) {  
3         int arr[] = {10, 50, 30, 40, 20};  
4  
5         int largest = arr[0];  
6         int second = arr[0];  
7  
8         for (int i = 1; i < arr.length; i++) {  
9             if (arr[i] > largest) {  
10                second = largest;  
11                largest = arr[i];  
12            } else if (arr[i] > second && arr[i] != largest) {  
13                second = arr[i];  
14            }  
15        }  
16  
17        System.out.println("Second Largest = " + second);  
18    }  
19 }
```

The IDE interface includes tabs for PROBLEMS, OUTPUT, DEBUG CONSOLE, TERMINAL, and PORTS. The TERMINAL tab is active, showing the command to run the program and its output:

```
PS D:\Java> & 'C:\Program Files\Android\Android Studio\jbr\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Sam...  
e\User\workspaceStorage\711c5833910ec7dcead394f26070c8f8\redhat.java\jdt_ws\Java_72da403b\bin' 'SecondLargest'  
Second Largest = 40  
PS D:\Java>
```